Abstract

This thesis details the development and characterization of an experimental setup for the controlled transport of cold ^{87}Rb atoms into a Hollow-Core Photonic Crystal Fiber (HCPCF) via an optical conveyor belt.
The initial stage of the work focuses on the preparation of the atomic sample. The optical molasses temperature was reduced to $T = (8.42 \pm 0.17)$ μK by tuning the cooling laser detuning and compensation magnetic fields. The loading sequence into the Optical Dipole Trap (ODT) was then investigated, reaching a capture efficiency of $(0.161 \pm 0.005)\%$ (corresponding to approximately 10^5 atoms) by optimizing the MOT position prior to the loading phase.
To implement the moving optical lattice, the injection of a 1064 nm beam into a polarization-maintaining single-mode fiber was designed and realized. The spatial overlap with the counter-propagating field emerging from the HCPCF was aligned on the upper breadboard. The stability of the transmitted light was evaluated using Allan-deviation analysis, showing a fractional power instability of 8×10^{-4} at 0.1 s and an angular polarization instability of 7×10^{-5} rad at 3 s.
Finally, since the transport sequence induces heating, a multi-level numerical model based on QuTiP was developed to simulate Electromagnetically Induced Transparency (EIT) cooling. The 24-level simulation indicates that a continuous repump field is necessary to mitigate optical pumping effects and sustain the cooling process toward the motional ground state.